

Math Review for ECON 630¹

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¹These slides are mainly based on Dr. Tian's note (You can find the note at <http://econweb.tamu.edu/tian/micro1.pdf>). Some materials are from the lectures I learned from Drs. Ning Sun, Rodrigo Velez, Yuzhe Zhang, as well as the references listed in Dr. Tian's note.

1.2.1 Sets, Functions

- Vector inequalities, $\geq$, $\geq$, and $>$ are defined as follows:

Let $a, b \in \mathbb{R}^n$, then

- $a \geq b$ means $a_s \geq b_s$ for all $s = 1, \dots, n$;
- $a \geq b$ means $a \geq b$ but $a \neq b$;
- $a > b$ means $a_s > b_s$ for all $s = 1, \dots, n$.

- Notation

- $\mathbb{R}$ is the set of real numbers.
- $\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_1 \geq 0, \dots, x_n \geq 0\}$, note that $x = (x_1, \dots, x_n)$.
- $\mathbb{R}_{++}^n = \{x \in \mathbb{R}^n : x_1 > 0, \dots, x_n > 0\}$.
- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are the sets of natural numbers, integers, and rational numbers.
- iff: if and only if.

1.2.1 Sets, Functions

Definition Let $X \subseteq \mathbb{R}^n$. A function $f : X \rightarrow \mathbb{R}$ is **continuous at a point** $x_0 \in X$ if

- $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, i.e.,
- $\forall \epsilon > 0, \exists \delta > 0$ such that for all $x \in X$ satisfying $|x - x_0| < \delta$ (i.e., $x \in B_\delta^X(x_0)$), we have $|f(x) - f(x_0)| < \epsilon$.

A function $f : X \rightarrow \mathbb{R}$ is **continuous on X** if it is continuous at every points in X .

Note that if $x, y \in \mathbb{R}^n$, $|x - y| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$.

1.2.1 Sets, Functions

Proposition If two functions f and g are continuous, then

- $f + g$, $f - g$, $f \cdot g$, and f/g ,
- the composition $f \circ g$,
- $\max\{f, g\}$

are continuous on their entire domains.

1.2.1 Sets, Functions

- Definition A set $C \subseteq \mathbb{R}^n$ is **closed** if every converging sequence $\{x^{(n)}\}_{n \in \mathbb{N}}$ in C converges to some point in C .
- Definition A set $X \subseteq \mathbb{R}^n$ is an **open** set if its complement is closed.

Example: the interval $[0, 1)$ is neither closed nor open in $\mathbb{R}$ since

- $\{1 - \frac{1}{n}\}_{n \in \mathbb{N}} \subseteq [0, 1)$, $1 - \frac{1}{n} \rightarrow 1$, but $1 \notin [0, 1)$;
 - if $[0, 1)$ is open in $\mathbb{R}$, then its complement $(-\infty, 0) \cup [1, \infty)$ should be closed, but this is not true.
- Proposition A set $X \subseteq \mathbb{R}^n$ is open iff for any $x \in X$ there exists some $\epsilon > 0$ such that $B_\epsilon^X(x) \subseteq X$, where $B_\epsilon^X(x) \equiv \{y \in X : |x - y| < \epsilon\}$.

1.2.1 Sets, Functions

- Prop. Meet of closed sets is closed; union of *finite* closed sets is closed.
- Prop. Union of open sets is open; meet of *finite* open sets is open.
- De Morgan's law: $\overline{A \cup B} = \overline{A} \cap \overline{B}$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$. The law also holds for arbitrary unions and intersections (meets).
- Example (meet of infinite open sets may not be open):
 $\{X_n = (-\frac{1}{n}, \frac{1}{n})\}_{n \in \mathbb{N}}$ are open sets. $\cap_{n=1}^{\infty} X_n = \{0\}$, not open!

1.2.1 Sets, Functions

- Definition Let $X \subseteq \mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$.
 - f is **upper semi-continuous** on X if the upper contour set is closed, *i.e.*, for each $\alpha \in \mathbb{R}$, $\{x \in X : f(x) \geq \alpha\}$ is closed in X .
 - f is **lower semi-continuous** if $-f$ is upper semi-continuous (*i.e.*, the lower contour set is closed).
- Proposition If f and g are two lower (upper) semi-continuous functions and $\lambda > 0$, then λf and $f + g$ are lower (upper) semi-continuous.

1.2.1 Sets, Functions

- Example Let $S = \{p \in \mathbb{R}_+^L : \sum_{l=1}^L p_l = 1\}$. f is a lower semi-continuous function satisfying $f(p) \cdot p \leq 0$ for all $p \in S$. Define

$$F(q) = \{p \in S : q \cdot f(p) \leq 0\} \quad \text{for all } q \in S.$$

Then,

- $F(q)$ is non-empty for each $q \in S$;
- $F(q)$ is closed for each $q \in S$.

1.2.1 Sets, Functions

- Definition A set $C \subseteq \mathbb{R}^n$ is called a **convex set** if for all $x, y \in C$ and all $\alpha \in [0, 1]$, we have $\alpha x + (1 - \alpha)y \in C$.
- Definition A set $C \subseteq \mathbb{R}^n$ is a **compact set** if C is both **bounded and closed**. C is **bounded** if there exists a finite number M such that for all $x \in C$, $|x| \leq M$.

1.2.1 Sets, Functions

- Let S be a set, not necessarily a subset of $\mathbb{R}^n$. Let $\mathcal{U}$ be a collection of open sets such that every point in S lies in at least one of the sets in $\mathcal{U}$: $S \subseteq \{U : U \in \mathcal{U}\}$. The collection $\mathcal{U}$ is called an **open cover** of S . The set S has the **finite covering property** if whenever $\mathcal{U}$ is an open cover of S , then a finite subcollection of the sets in $\mathcal{U}$ covers S ; that is, there exists $U_1, \dots, U_K \in \mathcal{U}$ such that $S \subseteq \bigcup_{k=1}^K U_k$.
- Theorem A set is a **compact set** iff it has the finite covering property.

1.2.1 Sets, Functions

- Weierstrass Theorem Any upper (lower) semi-continuous function reaches its maximum (minimum) on a **compact set**.

1.2.1 Sets, Functions

- Example 1.

$$F(x) = \begin{cases} -x & \text{if } x < 0 \\ x + 1 & \text{if } x \geq 0 \end{cases}$$

Is this function upper semi-continuous? Lower semi-continuous?

Reaches its maximum (minimum) on $[-1, 0]$?

- Example 2.

$$F(x) = \begin{cases} x & \text{if } x \leq 0 \\ -x + 1 & \text{if } x > 0 \end{cases}$$

Is this function upper semi-continuous? Lower semi-continuous?

Reaches its maximum (minimum) on $[0, 1]$?

1.2.2 Separating Hyperplane Theorem

- A **hyperplane** in $\mathbb{R}^n$ is a set $\{x \in \mathbb{R}^n : p \cdot x = c\}$, where $0 \neq p \in \mathbb{R}^n$ and $c \in \mathbb{R}$.
- Two sets A and B are said to be **strictly separated** by a hyperplane if there exists some nonzero $p \in \mathbb{R}^n$ and some $c \in \mathbb{R}$ such that for each $x \in A$ and each $y \in B$, $p \cdot x < c < p \cdot y$ (This is often written as $p \cdot A < c < p \cdot B$).

Note that for two vectors $a, b \in \mathbb{R}^n$, $a \cdot b$ is the inner product of them, i.e., $a \cdot b = a_1 * b_1 + \cdots + a_n * b_n$.

1.2.2 Separating Hyperplane Theorem

- **Separating Hyperplane Theorem** Suppose that $A, B \subseteq \mathbb{R}^n$ are **convex** and $A \cap B = \emptyset$. Then, there exists a vector $p \in \mathbb{R}^n$ with $p \neq 0$, and a number $c \in \mathbb{R}$ such that $\forall x \in A$ and $\forall y \in B$,
$$p \cdot x \leq c \leq p \cdot y.$$
- Furthermore, if A is **compact** and B is **closed**, there exists a vector $p \in \mathbb{R}^n$ with $p \neq 0$, and a number $c \in \mathbb{R}$ such that $\forall x \in A$ and $\forall y \in B$, $p \cdot x < c < p \cdot y$.

1.2.3 Concave and Convex Functions

- Definition Let X be a convex set in $\mathbb{R}^n$. A function $f : X \rightarrow \mathbb{R}$ is **concave** on X if for any $x, x' \in X$ and any t with $0 < t < 1$, we have $f(tx + (1 - t)x') \geq tf(x) + (1 - t)f(x')$. The function f is **strictly concave** on X if the inequality is strict for all $x \neq x'$.
- Definition A function f is (strictly) **convex** if $-f$ is (strictly) concave.
- Proposition $f : X \rightarrow \mathbb{R}$ is **convex** on X iff for any $x, x' \in X$ and any t with $0 < t < 1$, we have $f(tx + (1 - t)x') \leq tf(x) + (1 - t)f(x')$. f is **strictly convex** on X iff the inequality is strict for all $x \neq x'$.

1.2.3 Concave and Convex Functions

- Proposition A linear function is both concave and convex.
- Proposition The sum of two (strictly) concave (convex) functions is a (strictly) concave (convex) function.
- Proposition If f defined on $\mathbb{R}$ is twice differentiable, then f is concave (convex) **iff** $f'' \leq 0$ ($f'' \geq 0$).
- Proposition **If** $f'' < 0$ (> 0), then f is strictly concave (convex).
- Example: x^2 and x^4 are strictly convex function. $(x^2)'' = 2 > 0$, $(x^4)'' = 12x^2 \geq 0$ but not necessarily > 0 . $-x^2$ and $-x^4$ are strictly concave function

1.2.3 Concave and Convex Functions

- Proposition When a function f defined on a convex set $X \subseteq \mathbb{R}^n$ has continuous second partial derivatives,
 - f is concave (convex) **iff** the Hessian matrix $D^2 f(x)$ is negative (positive) semi-definite on X ;
 - f is strictly concave (strictly convex) **if** the Hessian matrix $D^2 f(x)$ is negative (positive) definite on X ;

where the Hessian is

$$D^2 f(x) = \left[\frac{\partial^2 f(x)}{\partial x_i \partial x_j} \right] = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}$$

1.2.3 Concave and Convex Functions

- Definition A $n \times n$ real symmetric matrix A is **negative definite** if $\forall x \neq 0, x^T A x < 0$. $\Leftrightarrow$ All k th order **leading principal minors** are < 0 if k is odd and > 0 if k is even.

The k -th ordered **leading principal minor** of A is a submatrix of A obtained when only the first k rows and columns are retained.

- Definition A is **negative semi-definite** if $\forall x \neq 0, x^T A x \leq 0$. $\Leftrightarrow$ All k th order **principal minors** are ≤ 0 if k is odd and ≥ 0 if k is even.

The k -th ordered **principal minor** of A is a submatrix of A when $n - k$ rows and the same $n - k$ columns are deleted.

1.2.3 Concave and Convex Functions

- Proposition When a function f defined on a convex set X is differentiable, then f is concave iff for any $x, x' \in X$,
$$f(x') \leq f(x) + Df(x)(x' - x), \text{ where } Df(x) = \left(\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right).$$
- Proposition If f is concave and increasing on $\mathbb{R}^1$ and g is concave, then the composite function $f \circ g$ is also concave, i.e., a increasing concave transformation of a concave function is concave.

1.2.3 Concave and Convex Functions

- Definition Let X be a convex set in $\mathbb{R}^n$. A function $f : X \rightarrow \mathbb{R}$ is **quasi-concave** on X if the upper contour set $\{x \in X : f(x) \geq \alpha\}$ is convex for all $\alpha \in \mathbb{R}$. It is **strictly quasi-concave** on X if the upper contour sets are strictly convex sets (a set C is strictly convex if for all $x, y \in C$, for all $t \in (0, 1)$, $tx + (1 - t)y$ belongs to the interior of C).
- Definition A function $f : X \rightarrow \mathbb{R}$ is **(strictly) quasi-convex** on X if $-f$ is (strictly) quasi-concave on X .
- Proposition $f : X \rightarrow \mathbb{R}$ is quasi-convex on X iff the lower contour set $\{x \in X : f(x) \leq \alpha\}$ is convex for all $\alpha \in \mathbb{R}$.

1.2.3 Concave and Convex Functions

- Proposition A function $f : X \rightarrow \mathbb{R}$ is quasi-concave on X iff for all $x, x' \in X$, and for all $t \in [0, 1]$, $f(tx + (1 - t)x') \geq \min\{f(x), f(x')\}$.
- Proposition A function $f : X \rightarrow \mathbb{R}$ is quasi-convex on X iff for all $x, x' \in X$, and for $t \in [0, 1]$, $f(tx + (1 - t)x') \leq \max\{f(x), f(x')\}$.
- Proposition (Strict) concavity/convexity $\implies$ (Strict) quasi-concavity/quasi-convexity.

1.2.3 Concave and Convex Functions

Proposition If f defined on a convex set $X(\subseteq \mathbb{R}^n)$ has continuous second partial derivatives. Define:

$$B_r(x) = \begin{vmatrix} 0 & f_1(x) & \cdots & f_r(x) \\ f_1(x) & f_{11}(x) & \cdots & f_{1r}(x) \\ \cdots & \cdots & \cdots & \cdots \\ f_r(x) & f_{r1}(x) & \cdots & f_{rr}(x) \end{vmatrix}$$

- a If f is quasi-concave, then $(-1)^r B_r(x) \geq 0$ for all $r = 1, \dots, n$;
- b If $(-1)^r B_r(x) > 0$ for all $r = 1, \dots, n$, then f is strictly quasi-concave;
- c If f is quasi-convex, then $B_r(x) \leq 0$ for all $r = 1, \dots, n$;
- d If $B_r(x) < 0$ for all $r = 1, \dots, n$, then f is strictly quasi-convex.

1.2.3 Concave and Convex Functions

- Proposition Quasiconcavity (NOT concavity) is preserved by monotonic transformation, i.e., if $f(\cdot)$ is (strictly) quasi-concave, g is (strictly) increasing, then $g \circ f$ is (strictly) quasiconcave.

Proof. $\forall x, y$ in the domain and $\forall \alpha \in (0, 1)$,

$$g(f(\alpha x + (1 - \alpha)y)) \geq g(\min\{f(x), f(y)\}) = \min\{g(f(x)), g(f(y))\}.$$

- Example $f(x) = \lambda_1 \ln x_1 + \cdots + \lambda_n \ln x_n$ defined on $\mathbb{R}_{++}^n$ with $\lambda_1, \cdots, \lambda_n > 0$ is strictly concave, and thus it is strictly quasi-concave. Its monotonic transformed function $u(x_1, \cdots, x_k) = Ax_1^{\lambda_1} \cdots x_n^{\lambda_n}$ (where $A > 0$) defined on $\mathbb{R}_{++}^n$ is quasi-concave, not concave when $\lambda_1 + \cdots + \lambda_n > 1$.

1.2.4 Optimization

Lagrangian Theorem (Equality constrained optimization)

$$\begin{aligned} \max \quad & f(x_1, \dots, x_n) \\ \text{s.t.} \quad & h_1(x_1, \dots, x_n) = d_1 \\ & \dots \\ & h_k(x_1, \dots, x_n) = d_k \end{aligned}$$

where $f, h_1, \dots, h_k$ are differentiable functions defined on $\mathbb{R}^n$ and $k < n$ and $d_1, \dots, d_k$ are constants. If the Jacobian matrix of the constraint functions, $\left[\frac{\partial h_i}{\partial x_j} \right]$ (a $k \times n$ matrix), has full row rank at an optimal solution. Let $L(x, \lambda) = f(x) - \sum_{i=1}^k \lambda_i [h_i(x) - d_i]$. Then the necessary conditions for the problem are the first-order conditions (FOC):

$$\frac{\partial L(x, \lambda)}{\partial x_i} = \frac{\partial f(x)}{\partial x_i} - \sum_{l=1}^k \lambda_l \frac{\partial h_l(x)}{\partial x_i} = 0 \quad i = 1, 2, \dots, n.$$

1.2.4 Optimization

Kuhn-Tucker Theorem (Inequality constrained optimization)

Consider the following optimization problem with equality constraints:

$$\begin{aligned} \max \quad & f(x_1, \dots, x_n) \\ \text{s.t.} \quad & h_i(x_1, \dots, x_n) \leq d_i \quad i = 1, 2, \dots, k. \end{aligned}$$

Suppose x is a local optimal solution for the optimization problem and satisfies the non-degenerate constraint qualification (NDCQ) condition.

Then, there are a set of Kuhn-Tucker multipliers $\lambda_1, \dots, \lambda_k \geq 0$ s.t.

$$\frac{\partial f(x)}{\partial x_i} = \sum_{l=1}^k \lambda_l \frac{\partial h_l(x)}{\partial x_i} \quad i = 1, 2, \dots, n.$$

$$\lambda_i \geq 0 \quad \text{for all } i = 1, 2, \dots, k.$$

$$\lambda_i = 0 \quad \text{if } g_i(x) < d_i.$$

1.2.4 Optimization

NDCQ Suppose x^* is a local optimal solution for the problem. If

- 1 gradient of all binding constraints at x^* are independent, *or*
- 2 constraints are linear, then, NDCQ is satisfied.

If NDCQ fails, then FOC may fail. Example:

$$\begin{aligned} \max \quad & x_1 \\ \text{s.t.} \quad & x_1^3 - x_2 \leq 0 \\ & x_2 \leq 0 \end{aligned}$$

The optimal solution is $(x_1^*, x_2^*) = (0, 0)$; both constraints are binding.

The gradients of bindings constraints at the optimal solution are

$(\frac{\partial h_1}{\partial x_1}, \frac{\partial h_1}{\partial x_2}) = (3x_1^{*2}, -1) = (0, -1)$, and $(\frac{\partial h_2}{\partial x_1}, \frac{\partial h_2}{\partial x_2}) = (0, 1)$. They are not independent. You can check that FOC does not hold.

1.2.4 Optimization

- **Kuhn-Tucker Theorem (Sufficiency 1)** Suppose f is concave and each g_i is convex. If x satisfies the Kuhn-Tucker first order conditions specified in the above theorem, then x is a global solution to the constrained optimization problem.
- **K-T Theorem (Sufficiency 2)** Suppose f is pseudo-concave and each g_i is quasi-convex. If x satisfies the Kuhn-Tucker first order conditions specified in the above theorem, then x is a global solution to the constrained optimization problem.
- **K-T Theorem (Sufficiency 3)** Suppose f is quasi-concave and the only one constraint is quasi-convex. If x satisfies the Kuhn-Tucker first order conditions specified in the above theorem, then x is a global solution to the constrained optimization problem.

1.2.4 Optimization

- Definition Let X be a convex subset of $\mathbb{R}^n$.

A function $f : X \rightarrow \mathbb{R}$ is **pseudo-concave** if for any $x, x' \in X$, $Df(x)(x' - x) \leq 0$ implies $f(x') \leq f(x)$.

- Proposition If f is concave, then f is pseudo-concave.
- Proposition If f is pseudo-concave, then f is quasi-concave.
- Proposition If X is open and $Df(x) \neq 0$ for all $x \in X$, then f is pseudo-concave iff f is quasi-concave.

1.2.4 Optimization: C-D utility function

$$\max u(x) = Ax_1^{\lambda_1} \cdots x_k^{\lambda_k}$$

$$\text{s.t. } p_1x_1 + \cdots + p_kx_k \leq m$$

$$x_1, \cdots, x_k \geq 0$$

p_i ($i = 1, \cdots, k$) are positive constants

- Is f concave (quasi-concave)? Is constraint convex (quasi-convex)?
- Forming the Lagrangian $L = Ax_1^{\lambda_1} \cdots x_k^{\lambda_k} - t(p_1x_1 + \cdots + p_kx_k - m)$
- First Order conditions:
- Optimal Solution: $x_i = \frac{\lambda_i}{\sum_{j=1}^k \lambda_j} \frac{m}{p_i}, \quad i = 1, \cdots, k.$

1.2.6 Point-to-Set Mappings

- When a mapping is not a single-valued function, but is a point-to-set mapping, it is called a **correspondence**. A correspondence F maps point x in the domain $X \subseteq \mathbb{R}^n$ into sets in the range $Y \subseteq \mathbb{R}^m$, and it is denoted by $F : X \rightarrow 2^Y$ or $F : X \rightarrow\rightarrow Y$.
- Definition A correspondence $F : X \rightarrow 2^Y$ is
 - (1) **non-empty valued** if the set $F(x)$ is non-empty for all $x \in X$;
 - (2) **closed valued** if the set $F(x)$ is a closed set for all $x \in X$;
 - (3) **compact valued** if the set $F(x)$ is a compact set for all $x \in X$.

1.2.6 Point-to-Set Mappings

- Definition A correspondence $F : X \rightarrow 2^Y$ is **upper hemi-continuous** (u.h.c.) at x if for each open set U containing $F(x)$, there is an open set $N(x)$ containing x such that if $x' \in N(x)$, then $F(x') \subseteq U$. A correspondence $F : X \rightarrow 2^Y$ is **upper hemi-continuous** if it is upper hemi-continuous at every $x \in X$.
- Remark: An open set U containing $F(x)$ can be viewed as a cap of $F(x)$ from above. By varying U , one can find tighter and tighter caps of $F(x)$. F is u.h.c. at x requires that each cap of $F(x)$ (no matter how tight it is) covers not only $F(x)$, but also $F(x')$ for all x' that are sufficiently close to x . Hence, $F(x')$ do not suddenly become much larger as x' moves away from x . Intuitively, F is u.h.c. at x implies F does not explode at x .

1.2.6 Point-to-Set Mappings

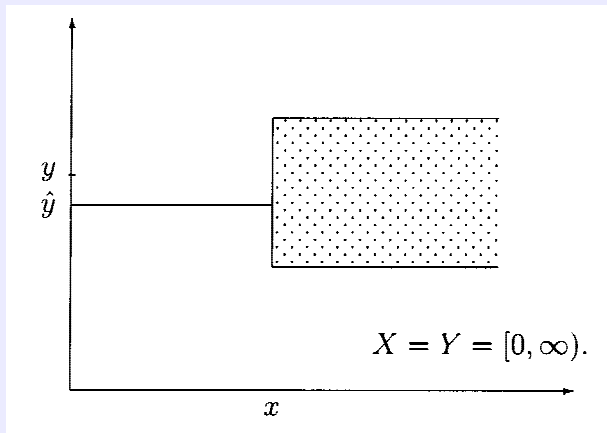


Figure: A u.h.c. (but not l.h.c.) correspondence.

1.2.6 Point-to-Set Mappings

- Definition A correspondence $F : X \rightarrow 2^Y$ is said to be **lower hemi-continuous** (l.h.c.) at x if for every open set V with $F(x) \cap V \neq \emptyset$, there exists a neighborhood $N(x)$ of x such that $F(x') \cap V \neq \emptyset$ for all $x' \in N(x)$. A correspondence $F : X \rightarrow 2^Y$ is **lower hemi-continuous** if it is lower hemi-continuous at every $x \in X$.
- Remark: An open set V with $F(x) \cap V \neq \emptyset$ can be viewed as a sensor of $F(x)$. By varying V , one can set sensors at every corner of $F(x)$. F is l.h.c. at x requires that each sensor of $F(x)$ catches not only $F(x)$, but also $F(x')$ for all x' that are sufficiently close to x . Hence, $F(x')$ do not suddenly disappear as x' moves away from x . Intuitively, F is l.h.c. at x implies F does not implode at x .

1.2.6 Point-to-Set Mappings

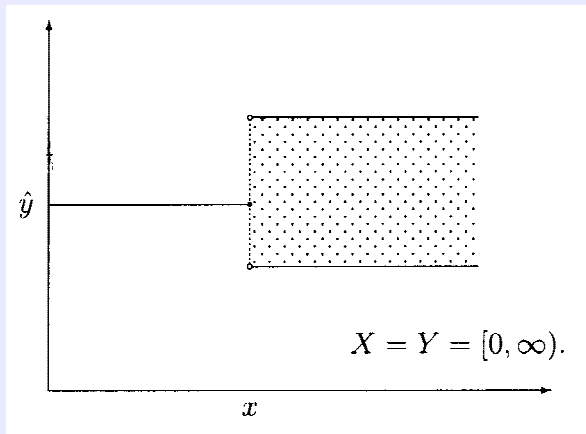


Figure: A l.h.c. (but not u.h.c.) correspondence.

1.2.6 Point-to-Set Mappings

- Proposition If a correspondence $F : X \rightarrow 2^Y$ is compacted-valued, then it is u.h.c. iff for any sequence $\{x_k\}$ with $x_k \rightarrow x$ and any $\{y_k\}$ with $y_k \in F(x_k)$, there exists a converging subsequence $\{y_{k_m}\}$ of $\{y_k\}$, $y_{k_m} \rightarrow y$, such that $y \in F(x)$.

1.2.6 Point-to-Set Mappings

- Definition A correspondence is said to be **continuous** if it is both upper hemi-continuous and lower hemi-continuous.
- Proposition A single-valued function is continuous as a function iff it is u.h.c. (or l.h.c.) as a correspondence.
- Proposition The budget constraint $B(p, I) = \{x \in \mathbb{R}_+^L : p \cdot x \leq I, p \in \mathbb{R}_{++}^L, I \in \mathbb{R}_{++}\}$ is both u.h.c. and l.h.c., i.e., it is continuous.

1.2.7 Continuity of a Maximum

- Berg's Maximum Theorem**

Suppose $f(x, a)$ is a **continuous function** mapping from $A \times X \rightarrow \mathbb{R}$, and the constraint set $F : A \rightarrow X$ is a **continuous correspondence** with **non-empty compact values**. Then, the optimal valued function:

$$M(a) = \max_{x \in F(a)} f(x, a)$$

is a continuous function in a ,

and the optimal solution correspondence:

$$\phi(a) = \arg \max_{x \in F(a)} f(x, a)$$

is compact-valued and upper hemi-continuous in a .

1.2.8 Fixed Point Theorems

- **Brouwer's Fixed Theorem**

Suppose that X is a non-empty subset of $\mathbb{R}^n$
and $f : X \rightarrow X$ is a function from X to X itself. If

- (a) X is compact and convex; and
- (b) f is continuous on X .

Then f has a fixed point, *i.e.*, there is a point $x^* \in X$ such that $f(x^*) = x^*$.

Brouwer's Fixed Theorem - Proof

- The unit L -simplex is defined as $\Delta^L \equiv \{x \in \mathbb{R}^{L+1} : \sum_{i=1}^{L+1} x_i = 1\}$.
- Let $e_l \in \mathbb{R}^{L+1}$ be a vector with l 's entry be 1 and all the others zero.
- Any $x \in \Delta^L$ is a convex combination of $e_1, e_2, \dots, e_{L+1}$.
- Sperner's labeling: for any point x in Δ^L , we label it as $i \in \{1, 2, \dots, L+1\}$ if i 's entry of x is non-zero.
- fully labeled: a sub-simplex (obtained from n -equilateral subdivision of Δ^L) is fully labeled if its vertexes are labeled $1, 2, \dots, L+1$.
- Sperner's lemma: the number of fully labeled sub-simplex for Sperner's labeling is odd.
- We can prove Sperner's lemma by induction and use Sperner's lemma to prove Brouwer's fixed point theorem.

1.2.8 Fixed Point Theorems

- **Kakutani's Fixed Point Theorem**

Suppose that X is a non-empty subset of $\mathbb{R}^m$ and $F : X \rightarrow \rightarrow X$ is a correspondence from X to X itself. If

- (a) X is compact and convex;
- (b) F is an upper hemi-continuous correspondences; and
- (c) $F(x)$ is convex for every $x \in X$.

Then there exists some $x^* \in X$ such that $x^* \in f(x^*)$.

Example: Symmetric Game

- Consider a two-person symmetric game satisfying the following conditions:
 - (a) for each player i , the strategy set S_i is a compact and convex subset of $\mathbb{R}^{L_i}$;
 - (b) for each player i , the payoff function $\Pi_i(s_i, s_{-i})$ is continuous in (s_i, s_{-i}) and quasi-concave in s_i ;
 - (c) the game is symmetric in the sense that $S_1 = S_2 = S$, and $\Pi_1(s, t) = \Pi_2(s, t)$ for all $s, t \in S$.
- Prove that the game has at least a symmetric Nash equilibrium.
 - 1. Best response correspondence $Br_1(s) = Br_2(s) = Br(s)$
 - 2. $Br(s)$ is u.h.c. (by Berg's theorem)
 - 3. $Br(s)$ is convex valued. (by quasi-concavity of $\Pi_i(s_i, s_{-i})$)
 - 4. Apply Kakutani's Fixed Point Theorem.

1.2.8 Fixed Point Theorems

• FKKM Theorem / KKM Lemma

Suppose that Y is a convex set and $\emptyset \neq X \subset Y$ and $F : X \rightarrow Y$ is a correspondence. If

- (a) $F(x)$ is closed for all $x \in X$;
- (b) $F(x_0)$ is compact for some $x_0 \in X$;
- (c) F is FS-convex, i.e., $\forall x_1, x_2, \dots, x_m \in X$ and its *convex combination* $(\sum_{i=1}^m \lambda_i = 1, \lambda_i \geq 0 \text{ for all } i = 1, \dots, m)$
 $x_\lambda = \sum_{i=1}^m \lambda_i x_i$, we have $x_\lambda \in \cup_{i=1}^m F(x_i)$.

Then $\cap_{x \in X} F(x) \neq \emptyset$.